Experiment 9: TCP Port Scan (Ports 20–1024)

Systems and Network Security Lab

Name: Karthik Anil
Roll No.: 20

1 T20 – SuperScan 4.1

1. Aim

To scan a given host for open TCP ports between 20 and 1024 and display each open port with its commonly associated service name using SuperScan 4.1 (T20), Nmap (T21), and Python sockets (T22).

2. Course Outcome and Module

Course Outcome: CO4.
Syllabus Program 9: Use SuperScan to scan a given host for open TCP ports between 20 and 1024 and display each open port with its service name. T20, T21, and T22 are three versions of this experiment.

3. Tools and Versions

Tool	Version / Environment
SuperScan	Version 4.1
Operating system	Windows 7 or later guest

4. Program or Commands

SuperScan is a graphical utility, so no separate source-code program is required. Use the GUI settings described below to configure the scan.

5. How to Run

1. Open the *Hostname/IP* tab and enter the authorised target in Start IP and End IP. Click > to add it to the list.
2. In *Host and Service Discovery*, enable TCP port scanning and set the range to 20–1024.
3. Open the *Scan* tab, press *Start*, then select *View Results*.
4. Record the open-port list and service names shown.

6. Sample Output with Screenshots

7. Result

SuperScan 4.1 was used to scan TCP ports 20–1024 on the configured target. The report showed one host discovered, zero open TCP ports, and zero open UDP ports.

Target: 127.0.0.1 Open TCP ports observed: 0

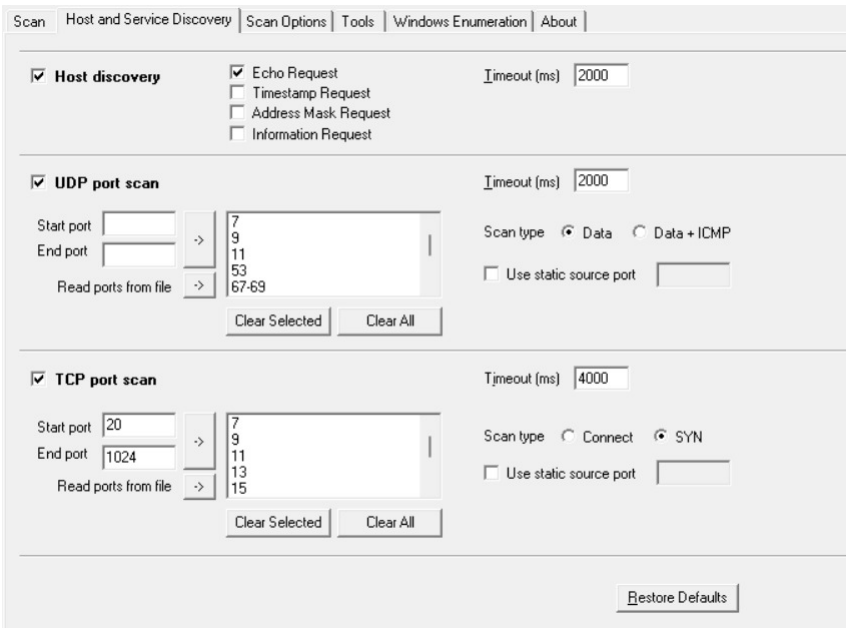


Figure 1: SuperScan 4.1 configuration for TCP port scanning.

SuperScan Report - 10/22/26 21:43:12

Total hosts discovered	1
Total open TCP ports	0
Total open UDP ports	0

Figure 2: SuperScan 4.1 TCP port scan results.

8. Difference from Other Versions

Version	Interface	Main distinction
T20 SuperScan	Graphical interface	Configure target and scan options in tabs
T21 Nmap	Command line	Uses Nmap scan options and service database
T22 Python	Python program	Uses sockets, threads, and configurable timeout

9. Viva: Five Questions with Answers

1. **What is a TCP port?** A numbered endpoint used by TCP network services.
2. **What is a port scan?** A check of selected ports to determine which appear open.
3. **Which port range is used?** TCP ports 20 through 1024.
4. **What is an open port?** A port that accepts a connection attempt.
5. **Why scan only authorised systems?** To avoid unauthorised access or disruption.

10. Common Errors: Message, Cause, Fix

Message	Cause	Fix
0 live hosts	Wrong IP or blocked discovery	Verify target IP and network connectivity
No open ports	No service listening or firewall filtering	Check the authorised target and active services
No results shown	Scan may not have started or completed	Confirm settings and use View Results

Overall Comparison

Task	Method	Service lookup	Distinct feature
T20	SuperScan GUI	Application display	Tab-based graphical workflow
T21	Nmap CLI	Nmap database	Flexible scan flags and port states
T22	Python sockets	`getservbyport()`	Threading and timeout controls

Academic Integrity Note

AI tools were utilized for grammatical review, comparison with alternative versions, assistance in preparing, reviewing, and verifying viva answers, and formatting this report. The results are from the student's practical run. All technical details should be checked against the lab instructions and the course's originality policy.